

A PulseWidth-IN-PulseWidth-Out Universal Nonlinear Processing Element for Time-Domain In-Memory Computing Systems

Yihao Chen^{1,3}, Pengcheng Feng^{1,3}, Zhigang Li^{1,3}, Gang Chen^{1,3}, Rongxuan Shen^{1,3}, Huaxiang Lu^{1,3*}, Xiaoxin Xu^{2,3}

¹Institute of Semiconductors, Chinese Academy of Sciences; ²Institute of Microelectronics, Chinese Academy of Sciences;

³University of Chinese Academy of Sciences.

*Corresponding author: luhx@semic.ac.cn

Abstract—Time-Domain In-Memory Computing (TD-IMC) has emerged as a promising analog computing architecture for edge AI applications. However, the lack of developed hardware operators, especially general nonlinear operators, necessitates frequent cross-domain data transmission in practical TD-IMC systems, significantly reducing energy efficiency. In this work, we propose a PulseWidth-IN-PulseWidth-OUT Universal Nonlinear Processing Element (PIPO-UNPE) to address the challenges of nonlinear processing in analog computing. By implementing an RRAM-based two-layer ReLU network, the PIPO-UNPE performs universal nonlinear operations entirely in the time domain. Algorithmically, we introduce Dynamic Loss-Responsive Subset Enhancement (DLRSE) to boost the performance of this low-cost network in function approximation tasks. From a hardware perspective, we design an RRAM-based pulse-driven programmable current source and a low-latency dispersion comparator-based voltage-to-time converter (VTC) to enhance both the energy efficiency and precision of the PIPO-UNPE. Hybrid simulations reveal that the PIPO-UNPE consumes 912 μ W of power while delivering a throughput of 10M NOPS (Nonlinear Operations Per Second). Incorporating the PIPO-UNPE into the TD-IMC accelerator can increase energy efficiency by a factor of 9.5 to 25, keeping the accuracy loss below 0.1%.

Index Terms—Time-Domain In-Memory Computing, Universal Nonlinear Processing Element, Neural Network Training, RRAM

I. INTRODUCTION

Time-Domain In-Memory Computing (TD-IMC), a promising computing architecture based on pulse-width modulation has been widely applied in Edge AI applications [1]. In TD-IMC systems, the multiply-accumulate (MAC) operations of Deep Neural Networks (DNNs) are executed in the analog domain with high energy efficiency, and the MAC results are represented and transmitted in the time domain using pulse widths. Compared to other IMC architectures, TD-IMC not only benefits from the improved time resolution offered by advanced process nodes, but also enables large-scale integration thanks to its simple data routing that only requires logic circuits [2].

Recent TD-IMC systems have primarily concentrated on optimizing the MAC macro. However, nonlinear operation in DNNs, such as the computations of the activation functions (AFs) and their derivatives (DAFs) during DNN inference and on-chip training respectively [3], have still relied on

digital processing elements (DPEs) for their complexity [4]–[7], necessitating the extensive use of power-hungry cross-domain modules like time-to-digital converters. To improve the energy efficiency of the TD-IMC, implementing an Analog Universal Nonlinear Processing Element (A-UNPE) for the direct nonlinear processing of pulse widths is essential. Notwithstanding, reported A-UNPEs [8]–[10], typically implemented by CMOS polynomial circuits, face challenges balancing high power consumption and limited support for various functions, stemming from the implementation of critical polynomial weights through digital-to-analog converters (DACs).

Instead of resorting to more complex polynomials, the ReLU neural network, which can be efficiently implemented in TD-IMC systems, has been proven to be an excellent universal function approximator [11]. An A-UNPE based on the ReLU neural network is expected to benefit from the inherent energy efficiency and scalability of the TD-IMC. Nonetheless, previous research has shown that simple ReLU networks struggle to achieve high approximation precision through naive backpropagation necessitating more complex network models and consequently increasing hardware costs [12].

In this work, the Dynamic Loss-Responsive Subset Enhancement (DLRSE) strategy is proposed to improve the approximation accuracy of the ReLU network. A PulseWidth-IN-PulseWidth-Out UNPE (PIPO-UNPE) based on the trained network is introduced. The key circuits of PIPO-UNPE include the MAC array constructed using RRAM-based Pulse-driven Programmable Current Sources (R-PPCS), as well as the Differential-IN-Differential-Out VTC (DIDO-VTC) and Single-IN-Differential-Out VTC (SIDO-VTC) implemented with low-latency dispersion comparators. Through hardware-software co-optimization, the PIPO-UNPE achieved an average precision of 9 bits when approximating multiple activation functions (ELU, GELU, Mish, Sigmoid, SiLU, and Tanh), while consuming 912 μ W of power and delivering a throughput of 10M NOPS. The simulation model was tested on the state-of-the-art TD-CIM architecture TIMELY [13] to evaluate the energy improvement and inference accuracy. The simulation results show that PIPO-UNPE can improve energy efficiency by factors ranging from 9.5 $\times$ to 25 $\times$ while incurring

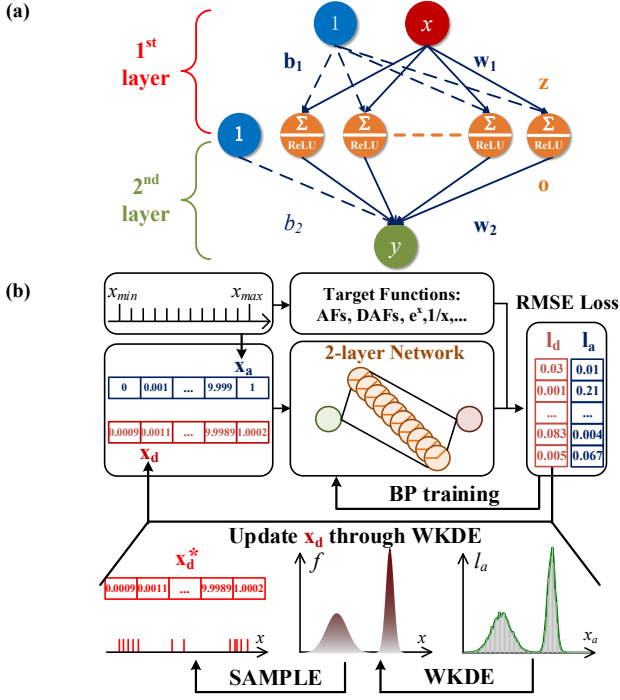


Fig. 1. (a) The proposed two-layer ReLU network for function approximation. (b) The completed training process using the DLRSE data augmentation strategy.

TABLE I
PERFORMANCE COMPARISON AND ABLATION STUDY RESULTS

Function	Domain	MAE ([12])	$R_{MAE} (N=16)$			$R_{MAE} (N=32)$		
			DLRSE	(w/o) x_d	(w/o) x_a	DLRSE	(w/o) x_d	(w/o) x_a
e^x	(-6,6)	0.096	1.4	2.8	3.3	0.6	0.9	1.6
$1/x$	(1,3)	0.0002	0.9	331.5	255.5	0.4	235.0	268.5
$1/\sqrt{x}$	(1,9)	0.0002	1.2	73.5	88.0	0.5	73.5	93.0
$1/(1+\sqrt{x})$	(1,9)	0.0001	0.9	48.2	55.6	0.3	45.3	53.1
Sigmoid	(-6,6)	0.0004	1.6	7.4	9.7	0.6	5.4	6.4
ELU	(-6,6)	0.0001	3.4	18.9	22.9	1.5	18.2	29.2
Tanh	(-6,6)	0.0006	1.8	5.1	7.2	0.6	3.0	3.7

¹ R_{MAE} is calculated as the ratio of the proposed model's MAE to the MAE of [12].

² The model in [12] contains $P = 5.793k$ parameters, whereas the proposed model has only $P = 48$ parameters when $N = 16$ and $P = 97$ parameters when $N = 32$.

an accuracy loss of less than 0.1%.

II. PROPOSED MODEL AND TRAINING STRATEGY

In our work, an IMC-friendly neural network, specifically a two-layer ReLU neural network, is embedded in the PIPO-UNPE as its core computational mechanism. As illustrated in Fig. 1(a), the first layer in the proposed model consists of N hidden neurons that execute an affine transformation $w_1x + b_1$ on the scalar input x , mapping it to an intermediate vector z . This vector is then transformed into a nonlinear vector o using the ReLU activation function. In the second layer, o undergoes a transformation $w_2o + b_2$, resulting in the approximated value y . Due to the limited effectiveness of naive

training strategies in achieving high precision for function approximation [12], [14], we introduce the Dynamic Loss-Responsive Subset Enhancement (DLRSE) data augmentation strategy, as depicted in Fig. 1(b). DLRSE comprises a static anchor dataset x_a and a dynamically updated dataset x_d . The x_a is obtained by uniformly selecting points within the fitting interval and remains unchanged during training, while x_d is updated every epoch based on the loss of x_a using Weighted Kernel Density Estimation (WKDE) to transform the loss into a probability density function, from which x_d^* is sampled. By exploiting the availability of labels in function approximation tasks, DLRSE enables the network to focus on high-loss regions while maintaining a comprehensive view of the function space. Furthermore, DLRSE augments the dataset in areas with high loss, allowing the network to gain a more complete understanding of the loss landscape during training, thereby facilitating the discovery of more optimal weights.

Table I demonstrates that, with 32 hidden neurons, the network trained with DLRSE achieves superior approximation accuracy compared to [12], while significantly reducing the number of parameters (P) to only 97 (64 in the first layer and 33 in the second layer), representing merely 1.67% of the parameters used in [12]. Ablation studies further validate the essential synergistic role of x_a and x_d in the success of DLRSE. It is noteworthy that in terms of weight efficiency ($\frac{1}{R_{MAE} \cdot P}$), the average efficiency for the network trained via conventional methods is approximately 1.7×10^{-4} [12]. A two-layer ReLU neural network trained by naive training strategies, (i.e., without x_d) achieves a similar value of 1.9×10^{-4} . In contrast, the proposed DLRSE-trained network attains a markedly higher value of 1.6×10^{-2} , attributable to the superior quality and diversity of training data dynamically supplied by DLRSE. This significant enhancement in weight efficiency indicates that the network can maintain equivalent approximation accuracy with a substantially more compact architecture, thereby reducing both the area and energy consumption of the nonlinear operator implemented using the trained network.

III. PROPOSED PIPO-UNPE

The schematics of the proposed PIPO-UNPE are shown in Fig. 2(a) and Fig. 2(b). The PIPO-UNPE comprises two cascaded layer circuits corresponding to the proposed two-layer network. In the neuron circuits of each layer, the R-PPCS array serves as the core unit responsible for storing weights and performing MAC operations. The implementation of R-PPCS is depicted in Fig. 2(c) and Fig. 2(d), which introduce two variants: R-PPCS-A and R-PPCS-B. Both variants consist of RRAM, a current buffer (CB), and current steering switches. The CB plays a critical role by providing the clamping voltage V_{BL} to read the RRAM conductance and transmitting the current I_G through the RRAM to the output of CB. The computation mechanism of R-PPCS is illustrated in Fig. 2(e) and Fig. 2(f). One R-PPCS can perform a one-quadrant (1Q) dot product. Subsequently, results from multiple R-PPCSs can be accumulated and stored by connecting them to the

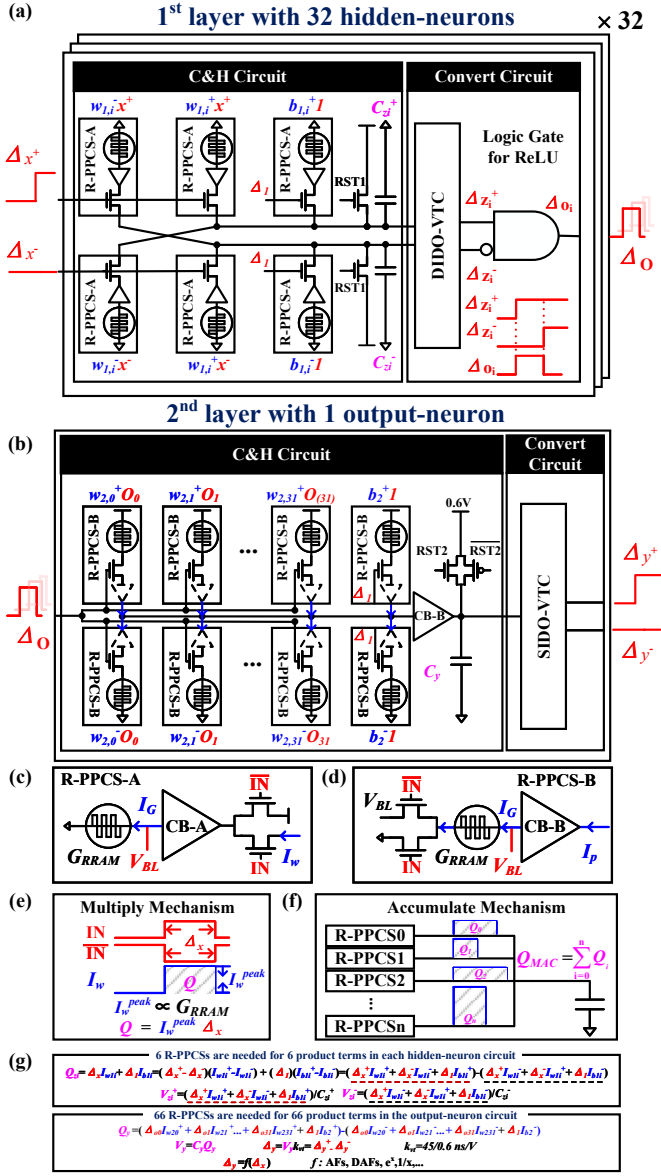


Fig. 2. Overview of the PIPO-UNPE System: (a)-(b) the circuit implementation of each layer in the proposed network, (c)-(d) diagrams of two types of R-PPCS, (e)-(f) the MAC mechanism and (g) the computation principles of each neuron circuit.

same capacitor. The difference between R-PPCS-A and R-PPCS-B lies in their array configurations. An R-PPCS-A array is composed of multiple parallel independent R-PPCS-A, whereas an R-PPCS-B array consists of multiple pairs of RRAM-switches sharing a single CB-B. Fig. 2(g) illustrates the computation principles of the PIPO-UNPE.

The functional waveform of the PIPO-UNPE is shown in Fig. 3. Both the hidden-neuron and output-neuron circuits follow an identical operation process, consisting of three phases: Charge & Hold (C&H), Convert, and Reset. During the C&H phase, the R-PPCSs charge the integration capacitors, driven by the input pulse width and the constant bias pulse width (Δ_1). In the Convert phase, computing results stored

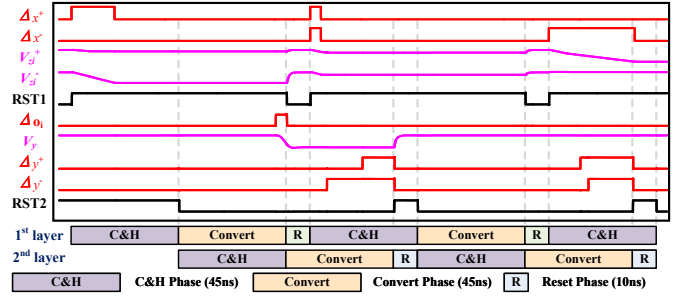


Fig. 3. The functional waveform of the PIPO-UNPE

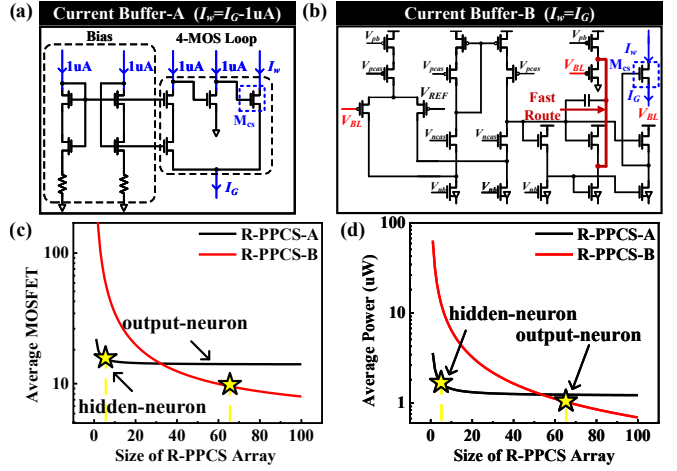


Fig. 4. (a)-(b) Circuit diagrams of CB-A and CB-B. (c) Average MOSFET overhead and (d) average power consumption for CB-A and CB-B arrays across various size.

in the integration capacitors are converted into differential pulse widths using the Differential-IN-Differential-Out VTC (DIDO-VTC) and the Single-IN-Differential-Out VTC (SIDO-VTC). The integration capacitors are reset during Reset phase to prepare for the next C&H phase. It is noteworthy that in order to achieve optimal computational accuracy and hardware efficiency, significant differences exist between the choice of R-PPCSs, bipolar large storage strategies, and VTC implementations in both hidden-neuron circuit and the output-neuron circuit due to their different scales of MAC operations. These distinctions will be discussed in detail subsequently.

A. R-PPCS-A VS R-PPCS-B

The primary distinction between R-PPCS-A of the hidden-neuron circuit and R-PPCS-B of the output-neuron circuit lies in their respective CBs. Fig. 4(a) and Fig. 4(b) illustrate the proposed CB-A and CB-B respectively. The main difference between CB-A and CB-B is the implementation of the M_{CS} gate voltage controller. CB-A features a four-transistor loop to ensure stable I_G transmission, whereas CB-B employs a two-stage amplifier with a fast response path to handle dynamic MAC current effectively. Although CB-B is individually more complex than CB-A, this does not imply that an R-PPCS-B array incurs higher hardware costs compared to an R-PPCS-A

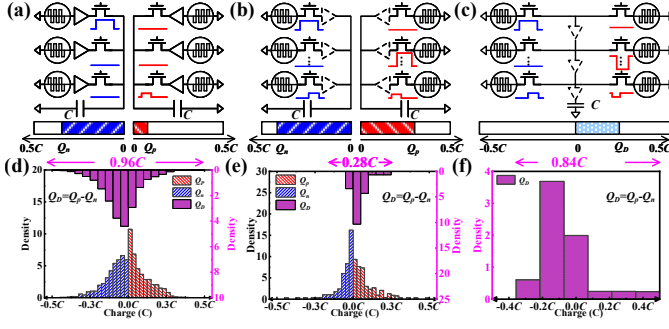


Fig. 5. (a)-(b) Differential capacitor pairs in the hidden-neuron and output-neuron circuits. (c) Pre-charge differentiation in the output-neuron circuit. (d)-(e) Overview of charge storage in the hidden and output-neuron circuits using differential capacitor pairs. (f) Situation of Q_D in the output-neuron circuit with pre-charge differentiation.

array, since multiple R-PPCS-Bs can share the same CB-B. As demonstrated in Fig. 4(c) and Fig. 4(d), simulation results indicate that R-PPCS-B is particularly well-suited for constructing large-scale R-PPCS arrays, which are necessary for the output-neuron circuit (sixty-six 1Q products). Conversely, R-PPCS-A is more appropriate for hidden-neuron circuits (six 1Q products) that require smaller-scale R-PPCS arrays.

B. Bipolar Charge Storage Method

Differential capacitor pairs are commonly used for bipolar charge storage in TD-IMC systems. As demonstrated in Fig. 5(a) and detailed in Fig. 5(d), the utilization of this technique within the hidden-neuron circuit yields a charge storage efficiency of approximately 92%. However, as shown in Fig. 5(b) and Fig. 5(e), this efficiency decreases to 48% within the output-neuron circuit due to common-mode charges ($Q_D = Q_p + Q_n$) being stored by the capacitors instead of differential-mode charges ($Q_C = Q_p - Q_n$), restricting the signal swing across capacitors and subsequently reducing output precision. To address this concern, as portrayed in Fig. 5(c), we propose to use opposing currents in the R-PPCS-B array within the output-neuron circuit to facilitate charge differentiation prior to reaching the capacitor, capitalizing on the CB-B's ability to transmit bidirectional currents. Following this modification, as demonstrated in Fig. 5(f), the storage efficiency significantly improves to reach 84%.

C. SIDO-VTC VS DIDO-VTC

Fig. 6(a) and Fig. 6(b) show the mechanisms of the DIDO-VTC and the SIDO-VTC. The DIDO-VTC converts differential voltages (V_z^+ and V_z^- , 0.6V-1.1V) to differential pulse widths; SIDO-VTC converts V_y (0.1V-1.1V) similarly. Fig. 6(c) depicts the ramp wave generator and comparator utilized by both VTCs. By employing constant-gm technology [15], we significantly reduce the conversion error (Δ_e), which is primarily caused by propagation delay dispersion between p-type and n-type transistors ($t_{PDp} - t_{PDn}$). The ramp generator's structure, analogous to the R-PPCS array, ensures stable slopes and enhances conversion accuracy. Fig. 6(d) and 6(e) present

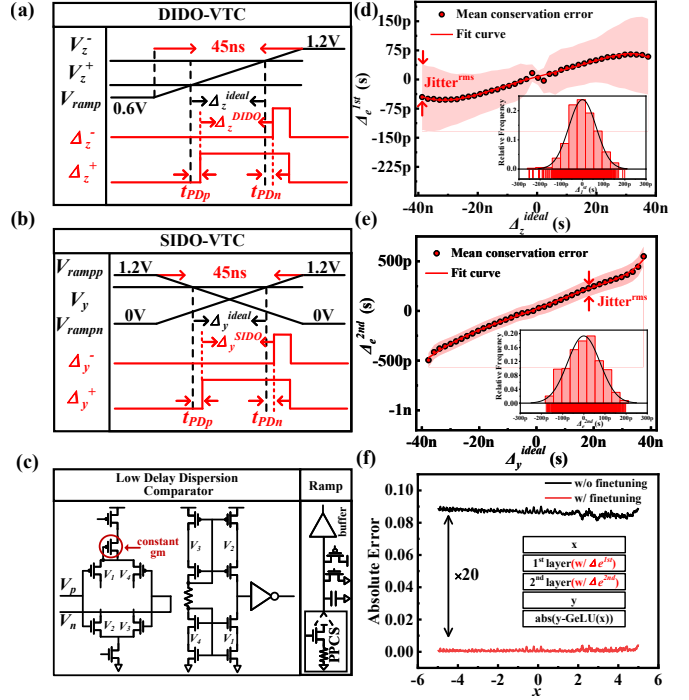


Fig. 6. (a)-(b) Operating mechanisms of DIDO-VTC and SIDO-VTC. (c) Circuit diagram of low delay dispersion comparator and ramp generator. (d)-(e) Conversion errors of DIDO-VTC and SIDO-VTC, where $\Delta_e^{1st} = \Delta_z^{ideal} - \Delta_z^{DIDO}$ and $\Delta_e^{2nd} = \Delta_y^{ideal} - \Delta_y^{SIDO}$. (f) Retrain the network to compensate for the precision loss introduced by VTCs.

the conversion errors under Monte Carlo simulations for the DIDO-VTC and SIDO-VTC, respectively. The conversion errors in both VTCs can be decomposed into two components: a systematic bias arising from inherent non-idealities in the circuit structure, manifested as the average conversion error, and random fluctuations due to process variations, resulting in jitter. While further reducing conversion errors can mitigate accuracy loss caused by circuit imperfections, compensating for these errors through traditional circuit design introduces additional hardware overhead. Alternatively, addressing this issue at the software level using hardware error-aware training strategies is more cost-effective. By leveraging the strong adaptability of neural networks, a network model further trained with hardware errors can effectively accommodate and compensate for these errors.

In this work, we focus on compensating for the relatively stable average conversion error, which dominates the overall error profile. Specifically, we performed a sixth-order polynomial approximation of the average conversion error and incorporated the approximated error curves of the first and second layers into the trained model as additive errors. As demonstrated in Fig. 6(f), using the GeLU function as an example, the approximation accuracy of the PIPO-UNPE improves nearly 20-fold after fine-tuning compared to its pre-tuned performance.

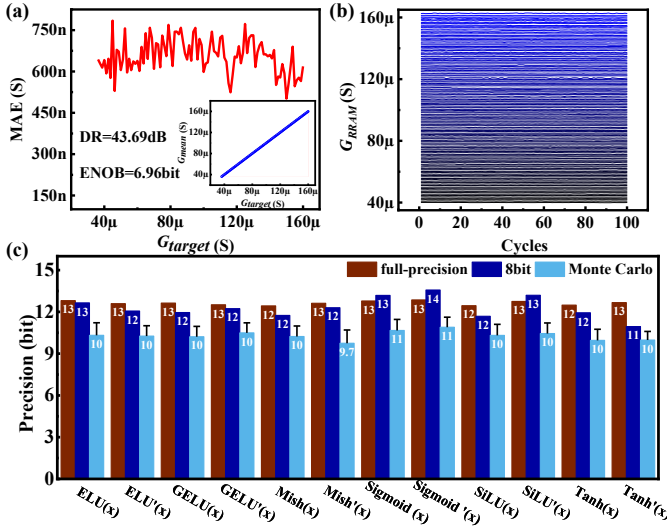


Fig. 7. (a) Random target programming results for 512 RRAMs. (b) The relaxation of 128 programmed states, in which each conductance level provided by 20 random RRAMs. (c) Approximation precision of PIPO-UNPE under hybrid simulation, calculated as $(20\lg(10/\text{MAE})-1.76)/6.02$.

IV. RESULTS AND DISCUSSION

To bridge the performance gap between simulation and hardware, we adopted a hybrid simulation strategy in our study. The RRAM conductance values for the PIPO-UNPE were sourced from a commercial RRAM array, and the circuit design and simulation were based on a 130 nm technology library. For device performance evaluation, we conducted random programming experiments to explore the continuous programming characteristics of the RRAM in PIPO-UNPE. As shown in Fig. 7(a), a total of 512 valid RRAMs were randomly programmed to target conductance values within a precision requirement of $\pm 0.5\%$ across the conductance range from 40 μS to 160 μS . The RRAM exhibited a dynamic range (DR) of 43 dB, corresponding to approximately 7 bits of accuracy. Utilizing the results from the random programming experiments, we programmed the RRAMs linearly into 128 conductance states. As shown in Fig. 7(b), the conductance states after programming exhibited minimal overlap over 100 read cycles, indicating excellent state stability within the selected programming range, where the RRAM operates in the high-conductance state. In subsequent experiments, the measured RRAM conductance data were directly applied to the circuit simulation to reliably evaluate the performance of

TABLE II
MODEL ACCURACY WITH DIFFERENT ACTIVATION FUNCTIONS

Quantization	MobileViT	ConvNeXt	YOLOv8	EfficientNetV2
8-bit Weight FP Activation	77.7%	82.4%	48.5%	83.2%
8-bit Weight 8-bit Activation	76.1%	81.1%	47.2%	82.4%
8-bit Weight PIPO-UNPE	77.4%	82.2%	48.7%	83.1%

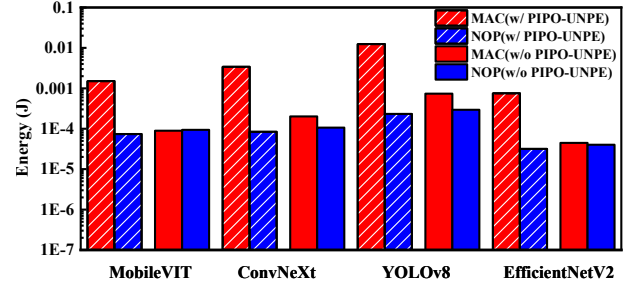


Fig. 8. Energy efficiency enhancements of TIMELY caused by PIPO-UNPE, with the digital nonlinear module utilizing NN-LUT scaled down to 130 nm [5].

PIPO-UNPE. For function testing, we designed approximation tasks for six types of activation functions and their derivatives, totaling twelve functions, to assess the nonlinear computation performance of PIPO-UNPE. Specifically, for each function, we first trained the two-layer neural network embedded within PIPO-UNPE using DLRSE to approximate the target function. Then, we quantized the trained full-precision model to 8 bits to align with the 7-bit maximum precision achievable by a single RRAM cell. Finally, the conductance data sequences corresponding linearly to the quantized weights were loaded into the circuit model for Monte Carlo simulation testing. As illustrated in Fig. 7(c), PIPO-UNPE achieved an average precision of 9.5 bits, representing a 3.4 bits precision loss compared to the full-precision model.

Notably, when applied to advanced network models, PIPO-UNPE meets the computational requirements of the activation functions used in these models. As shown in Table II, we tested four models—MobileViT, ConvNeXt, YOLOv8, and EfficientNetV2—using full-precision (FP) activation functions, 8-bit quantized activation functions, and activation functions implemented with the proposed PIPO-UNPE. The experimental results indicate that PIPO-UNPE performs comparably to full-precision activation functions, with an average accuracy loss of only 0.1%, which is significantly lower than the 1.25% degradation observed with 8-bit quantized activation functions.

Furthermore, integrating PIPO-UNPE into TD-CIM accelerators can significantly enhance their energy efficiency. For instance, in the TD-CIM accelerator TIMELY [13], the absence of nonlinear pulse-width computing resources necessitates transmitting the pulse widths output by the MAC array to the digital domain via TDCs to perform all computations required by the model. In contrast, PIPO-UNPE enables direct nonlinear processing of pulse widths, allowing a substantial portion of the MAC outputs to remain in the analog domain and thereby reducing the need for digital-to-analog conversions. As illustrated in Fig. 8, incorporating PIPO-UNPE improves the energy efficiency of TIMELY by factors of 9.5 $\times$, 19 $\times$, 25 $\times$, and 12 $\times$ for MobileViT, ConvNeXt, YOLOv8, and EfficientNetV2, respectively.

The power and area breakdowns of the PIPO-UNPE design are shown in Fig. 9. The total power consumption of PIPO-UNPE is 912 μW , and the total area is 10,516 μm^2 . Table

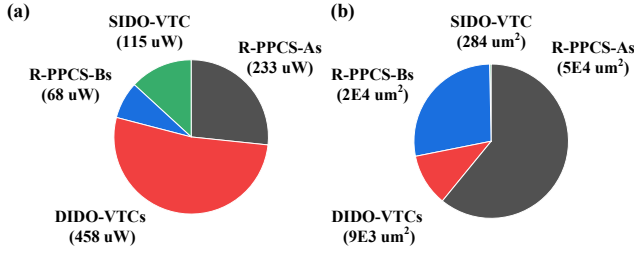


Fig. 9. Breakdown of the PIPO-UNPE: (a) power and (b) area.

TABLE III
COMPARISON WITH PREVIOUS DIGITAL AND ANALOG UNPE

	GEBA [4]	NN-LUT [5]	NACU [6]	ReAFM [7]	RAM-NAF [8]	PIPO-UNPE
Technology (nm)	28	7	28	20	65	130
Operation Mode	Digital	Digital	Digital	Digital	Time	Time
Functions (#)	4	4	5	10	3	12
Throughput (NOPS)	12.5M	360M	33M	40M	78M	10M
Power (mW)	9	0.025	1.035	N/A	0.168	0.912
Energy Efficiency (GNOPS/W)	1.4	1440	31	N/A	464	11
Normalized EE [†] (GNOPS/W)	0.71	8.1	2.1	N/A	7.2	11
Shareable	Yes	Yes	Yes	Yes	No	Yes
Reallocatable	No	No	No	No	No	Yes

[†] Normalized Energy Efficiency (EE): Power and throughput are scaled to a 130 nm process node, incorporating the TDC and DAC required for UNPE operations to ensure accurate EE evaluation [1], [16].

III presents a comparison table of PIPO-UNPE with other digital-analog UNPEs. Notably, after normalizing for size, PIPO-UNPE demonstrates a significant advantage in energy efficiency. Moreover, compared to other analog nonlinear operation schemes PIPO-UNPE offers two main advantages. First, it directly processes pulse-width signals that are easy to route, making it shareable among MAC arrays. Second, unlike traditional analog UNPEs that rely on polynomial approximations, the neural network-based PIPO-UNPE allows flexible reallocation of its computational resources as linear computation resources. This enables TD-CIM accelerators to dynamically distribute computational resources between linear and nonlinear operations according to the model's requirements.

V. CONCLUSION

In this work, we propose the PIPO-UNPE to address the lack of universal analog nonlinear operators in TD-IMC systems. The DLRSE effectively resolves the issue of low weight efficiency in traditional neural network training methods. Simulation results demonstrate that PIPO-UNPE not only exhibits high performance but also significantly enhances the energy efficiency of existing accelerators. Notably, because the computational resources of PIPO-UNPE align with those of existing accelerators, its integration is seamless and scalable—features not available in other proposed Nonlinear Processing Elements. We believe that PIPO-UNPE will strongly

promote the development of flexible and computationally powerful TD-IMC accelerators.

ACKNOWLEDGMENT

This work was supported in part by the National Natural Science Foundation of China under Grant 92364202 and Grant 62322412.

REFERENCES

- [1] J.-M. Hung, T.-H. Wen, Y.-H. Huang, S.-P. Huang, F.-C. Chang, C.-I. Su, W.-S. Khwa, C.-C. Lo, R.-S. Liu, C.-C. Hsieh *et al.*, “8-b precision 8-mb rram compute-in-memory macro using direct-current-free time-domain readout scheme for ai edge devices,” *IEEE Journal of Solid-State Circuits*, vol. 58, no. 1, pp. 303–315, 2022.
- [2] A. Whitcombe, S. Kundu, H. Chandrakumar, A. Agrawal, T. Brown, S. Callender, B. Carlton, and S. Pellerano, “22.3 a 76mw 40gs/s 7b time-interleaved hybrid voltage/time-domain adc with common-mode input tracking,” in *2024 IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 67. IEEE, 2024, pp. 392–394.
- [3] C. Lu, B. Shi, and L. Chen, “Analogue circuit realization of a programmable sigmoidal function and its derivative for on-chip bp learning,” in *IEEE APCCAS 2000. 2000 IEEE Asia-Pacific Conference on Circuits and Systems. Electronic Communication Systems (Cat. No. 00EX394)*. IEEE, 2000, pp. 626–629.
- [4] C. Ye and D. S. Jeong, “Geba: Gradient-error-based approximation of activation functions,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, 2023.
- [5] J. Yu, J. Park, S. Park, M. Kim, S. Lee, D. H. Lee, and J. Choi, “Nn-lut: neural approximation of non-linear operations for efficient transformer inference,” in *Proceedings of the 59th ACM/IEEE Design Automation Conference*, 2022, pp. 577–582.
- [6] G. Baccelli, D. Stathis, A. Hemani, and M. Martina, “Nacu: a non-linear arithmetic unit for neural networks,” in *2020 57th ACM/IEEE Design Automation Conference (DAC)*. IEEE, 2020, pp. 1–6.
- [7] X. Wu, S. Liang, M. Wang, and Z. Wang, “Reafm: A reconfigurable nonlinear activation function module for neural networks,” *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 70, no. 7, pp. 2660–2664, 2023.
- [8] C. Xie, Z. Shao, Y. Wu, C. Liu, M. Zhang, Y. Du, and L. Du, “Ram-naf: A reconfigurable analog module for nonlinear activation function computation in processing-in-memory,” *Authorea Preprints*, 2024.
- [9] S. Vlassis, F. Khateb, and G. Souliotis, “An on-chip linear, squaring, cubic and exponential analog function generator,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 66, no. 1, pp. 94–104, 2018.
- [10] M. T. Abuelma’ati and A. Shwehneh, “A reconfigurable satlin/sigmoid/gaussian/triangular basis functions computation circuit,” in *APCCAS 2006-2006 IEEE Asia Pacific Conference on Circuits and Systems*. IEEE, 2006, pp. 1172–1175.
- [11] I. Daubechies, R. DeVore, S. Foucart, B. Hanin, and G. Petrova, “Nonlinear approximation and (deep) relu networks,” *Constructive Approximation*, vol. 55, no. 1, pp. 127–172, 2022.
- [12] X. Xu, Z. Jiang, G. Chen, Z. Li, G. Gong, and H. Lu, “General nonlinear function neural network fitting algorithm based on cnn,” in *2022 3rd International Conference on Computer Vision, Image and Deep Learning & International Conference on Computer Engineering and Applications (CVIDL & ICCEA)*. IEEE, 2022, pp. 1079–1082.
- [13] W. Li, P. Xu, Y. Zhao, H. Li, Y. Xie, and Y. Lin, “Timely: Pushing data movements and interfaces in pim accelerators towards local and in time domain,” in *2020 ACM/IEEE 47th Annual International Symposium on Computer Architecture (ISCA)*. IEEE, 2020, pp. 832–845.
- [14] B. Liu and Y. Liang, “Optimal function approximation with relu neural networks,” *Neurocomputing*, vol. 435, pp. 216–227, 2021.
- [15] M. Wang, T. L. Mayhugh, S. H. Embabi, and E. Sánchez-Sinencio, “Constant-g/sub m/rail-to-rail cmos op-amp input stage with overlapped transition regions,” *IEEE Journal of Solid-State Circuits*, vol. 34, no. 2, pp. 148–156, 1999.
- [16] Z. Xuan and Y. Kang, “High-efficiency data conversion interface for reconfigurable function-in-memory computing,” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 30, no. 9, pp. 1193–1206, 2022.